

Equal powers of one base

Assume $b > 0$ and $b \neq 1$; u and v are real.

$$b^u = b^v$$

When may we conclude that $u = v$?

Like bases

$$3^{2x-1} = 27$$

What is the exact solution?

Rewriting a base

$$4^{x+1} = 2^{3x}$$

What is the exact solution?

A reciprocal base

$$5^{x-2} = \frac{1}{25}$$

What is the exact solution?

Exponents on both sides

$$2^{3x-1} = \frac{2^{x+2}}{4}$$

What is the exact solution?

Isolating an exponential

$$4 \cdot 3^x - 5 = 31$$

What is the exact solution?

An impossible real output

$$2^x = -7$$

Does this equation have a real solution?

Using a logarithm

$$5^x = 17$$

What is the exact solution expressed with natural logarithms?

An exact value and its approximation

$$2^{3x} = 11$$

What is the exact solution and its approximation to four decimal places?

A shifted exponent

$$7^{x-2} = 20$$

What is the exact solution?

Different bases

$$3^x = 5^{x-1}$$

What is the exact solution?

A natural exponential

$$6e^{0.3x} = 42$$

What is the exact solution?

A quadratic in an exponential

$$4^x - 5 \cdot 2^x + 4 = 0$$

What are all real solutions?

Definition of a logarithm

$$\log_3 (x - 2) = 2$$

What solution belongs to the original domain?

A natural logarithmic equation

$$\ln(2x + 1) = 3$$

What is the exact solution?

The one-to-one property

Assume $b > 0$, $b \neq 1$, $M > 0$, and $N > 0$.

$$\log_b M = \log_b N$$

When may we conclude that $M = N$?

Equal logarithms

$$\ln(3x + 1) = \ln(x + 7)$$

What solution satisfies both argument restrictions?

Condensing before solving

$$\log_2 x + \log_2 (x - 2) = 3$$

What solution survives the original domain restrictions?

A logarithmic difference

$$\ln(x + 2) - \ln x = \ln 3$$

What solution satisfies the original equation?

A logarithm with no real solution

$$\ln(x - 1) = \ln(1 - x)$$

Can any real x make both sides defined?

Checking candidate solutions

$$\ln(x^2 - 3x) = \ln 4$$

Which candidates satisfy the original argument restriction?

A growth threshold

P is a population count and t is elapsed years, $t \geq 0$. Assume the model holds continuously.

$$P(t) = 500(1.08)^t$$

When does the model first reach 900, to the nearest tenth of a year?

A decay threshold

M is mass in grams and t is elapsed hours, $t \geq 0$.

$$M(t) = 80e^{-0.15t}$$

When does the sample reach 20 grams, to the nearest tenth of an hour?

Doubling under continuous interest

An account with a positive initial balance earns 4% annual interest compounded continuously, with no deposits, withdrawals, or fees.

How many years does doubling take, rounded to the nearest tenth?